Microeconomics

Lecture 3 · Duality and the Slutsky equation

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What we know so far

- Preferences and utility: how to represent what a consumer wants.
- Budgets and utility maximisation: what the consumer can choose.
- Marshallian demand, corners and Kuhn–Tucker conditions.
- Indirect utility and the envelope theorem.

What we will learn today

- Expenditure minimisation and Hicksian demand.
- Duality: two ways to describe the same optimal choice.
- Shephard’s lemma and compensated price responses.
- Substitution and income effects: the Slutsky equation.

What happens when prices change?

We want to understand how a price change affects our choices.

Relative prices change: some goods become more expensive compared with others.

Purchasing power changes: the same income buys a different standard of living.

We need a way to separate these two effects.

The expenditure minimisation problem

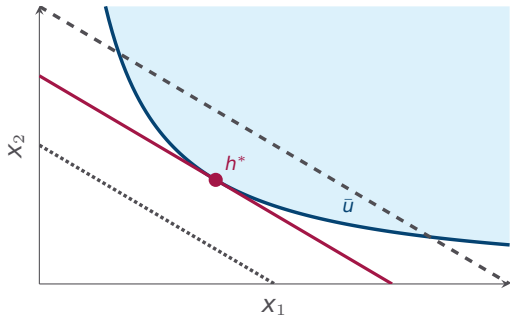
$$\min_{x} \quad p \cdot x$$

$$\text{subject to} \qquad u(x) \geq \bar{u}, \qquad x \geq 0.$$

$$e(p, \bar{u}) = \min_{x \geq 0,\ u(x) \geq \bar{u}} p \cdot x, \qquad H(p, \bar{u}) = \arg\min_{x \geq 0,\ u(x) \geq \bar{u}} p \cdot x.$$

e: expenditure function. H: Hicksian demand correspondence; h if unique.

Reach a target indifference curve at minimum cost



Start with a target indifference curve.
The shaded region gives at least this utility.
A higher spending line reaches the target.
A lower spending line cannot reach it.
Minimum spending: the cheapest line that reaches the target.

The assumptions we will use

- Consumption set $\mathbb{R}^n_+$; prices $p \gg 0$.
- u is continuous. The target $\bar{u} \geq u(0)$ is attainable.
- Add locally nonsatiated preferences for the duality results.
- Add uniqueness and smoothness only where derivatives require them.

Does a minimum exist?

Proposition. If u is continuous, $p \gg 0$ and $\bar{u}$ is attainable, then $H(p, \bar{u}) \neq \varnothing$.

The feasible upper contour set may be unbounded.

How can we still use a compactness argument?

Find *one* feasible bundle and discard every bundle that costs more.

Existence: cut away expensive bundles

Choose $\widehat{x} \geq 0$ with $u(\widehat{x}) \geq \bar{u}$. Set

$$K = \{x \geq 0 : u(x) \geq \bar{u},\ p \cdot x \leq p \cdot \widehat{x}\}.$$

K is nonempty and closed. Positive prices make it bounded.

The continuous function $p \cdot x$ attains a minimum on compact K.

Every excluded feasible bundle costs more than $p \cdot \widehat{x}$, so this is a global minimum.

No excess utility

Proposition. Under the basic assumptions, every $h \in H(p, \bar{u})$ satisfies $u(h) = \bar{u}$.

Suppose $u(h) > \bar{u}$. If $h = 0$, this contradicts $\bar{u} \geq u(0)$.

Otherwise, continuity gives $t < 1$ close to 1 with $u(th) > \bar{u}$.

But $p \cdot (th) < p \cdot h$: a cheaper feasible bundle. Contradiction.

Change the currency, not the chosen bundle

For every $t > 0$:

$$H(tp, \bar{u}) = H(p, \bar{u}), \qquad e(tp, \bar{u}) = t\, e(p, \bar{u}).$$

Proof. The constraint $u(x) \geq \bar{u}$ does not change.

Multiplying every feasible bundle's cost by t preserves all cost rankings.

Hicksian demand has degree 0 in prices; expenditure has degree 1.

Convex preferences and Hicksian choices

Convex preferences imply a convex set $H(p, \bar{u})$.

If $h, h' \in H$, their mixture stays target-feasible and costs exactly $e(p, \bar{u})$.

Strictly convex preferences imply a unique Hicksian choice.

Two distinct minimisers have utility exactly $\bar{u}$. Their mixture would give more than $\bar{u}$, violating no excess utility.

Buy a Cobb–Douglas standard of living

$$u(x) = \sqrt{x_1 x_2}, \qquad p \gg 0, \quad \bar{u} > 0.$$

Find Hicksian demand and the minimum expenditure needed to reach $\bar{u}$.

Then evaluate your answer at $p = (4, 1)$ and $\bar{u} = 3$.

Cobb–Douglas: the dual solution

Minimise cost while attaining the target $u(x) \geq \bar{u} > 0$.

$$h_1 h_2 = \bar{u}^2, \qquad p_1 h_1 = p_2 h_2.$$

$$h_1 = \bar{u}\sqrt{p_2/p_1}, \qquad h_2 = \bar{u}\sqrt{p_1/p_2}.$$

Thus $e(p, \bar{u}) = 2\bar{u}\sqrt{p_1 p_2}$.

At $p = (4, 1)$ and $\bar{u} = 3$: $h = (3/2, 6)$ and $e = 12$.

Perfect complements: find the cheapest bundle

$$u(x) = \min\{x_1, x_2\}, \qquad p \gg 0, \quad \bar{u} \geq 0.$$

Find Hicksian demand and expenditure.

The target is equivalent to $x_1 \geq \bar{u}$ and $x_2 \geq \bar{u}$.

$$h(p, \bar{u}) = (\bar{u}, \bar{u}), \qquad e(p, \bar{u}) = (p_1 + p_2)\bar{u}.$$

The direct utility has a kink, but expenditure is smooth in prices.

Perfect substitutes: which route is cheapest?

For $u(x) = x_1 + x_2$, $p \gg 0$ and $\bar{u} > 0$, solve

$$\min_{x \geq 0} \; p_1 x_1 + p_2 x_2 \qquad \text{subject to } x_1 + x_2 \geq \bar{u}.$$

Find expenditure and all minimisers, including the case of equal prices.

$$e(p, \bar{u}) = \bar{u} \min\{p_1, p_2\}.$$

$p_1 < p_2$: $h = (\bar{u}, 0)$. $\qquad$ $p_1 > p_2$: $h = (0, \bar{u})$.

$p_1 = p_2$: $H = \{x \geq 0 : x_1 + x_2 = \bar{u}\}$.

Two problems, the same boundary

Assume continuous LNS utility, $p \gg 0$, $m > 0$, and an attainable $\bar{u} > u(0)$.

$$e(p, v(p, m)) = m, \qquad v(p, e(p, \bar{u})) = \bar{u}.$$

$$X(p, m) = H(p, v(p, m)), \qquad H(p, \bar{u}) = X(p, e(p, \bar{u})).$$

The equalities hold for *sets of choices*, including nonunique optima.

Proof: the cheapest way to buy the maximum

Choose $x^* \in X(p, m)$ and let $\bar{u} = v(p, m)$. Walras gives $p \cdot x^* = m$.

x^* attains $\bar{u}$, so $e(p, \bar{u}) \leq m$.

If $e(p, \bar{u}) < m$, a minimiser h attains at least $\bar{u}$ at cost below m.

Then h is also utility-maximising with a slack budget, contradicting LNS and Walras. Thus $e(p, v(p, m)) = m$.

Proof: the most utility the minimum cost buys

Choose $h \in H(p, \bar{u})$ and set $m = e(p, \bar{u})$. No excess utility gives $u(h) = \bar{u}$.

Since h is affordable, $v(p, m) \geq \bar{u}$.

If an affordable z gave $u(z) > \bar{u}$, then z would also minimise expenditure for $\bar{u}$.

That violates no excess utility. Hence $v(p, e(p, \bar{u})) = \bar{u}$.

Proof: match the sets, not just their values

$$X(p, m) = \{x \geq 0 : u(x) \geq v(p, m),\ p \cdot x \leq m\}.$$

$$H(p, \bar{u}) = \{x \geq 0 : u(x) \geq \bar{u},\ p \cdot x \leq e(p, \bar{u})\}.$$

Substitute $\bar{u} = v(p, m)$ and $e(p, v(p, m)) = m$: the right-hand sides coincide.

Substitute $m = e(p, \bar{u})$ for the other equality.

Check duality for Cobb–Douglas

Let $u(x) = \sqrt{x_1 x_2}$, with $p \gg 0$, $m > 0$ and $\bar{u} > 0$.

Find $v(p, m)$, $e(p, \bar{u})$ and the two demand functions.

Verify directly that

$$e(p, v(p, m)) = m, \qquad v(p, e(p, \bar{u})) = \bar{u}.$$

Verify that $h(p, v(p, m)) = x(p, m)$.

Checkpoint

- Every price doubles; the utility target stays fixed. What happens to Hicksian quantities and expenditure?
- We relabel utility as $\widetilde{u} = 2u + 10$. Which new target represents the old level $u = 7$?
- If utility maximisation has two optimal bundles, can the corresponding expenditure problem have a unique solution?

Shephard's lemma

Fix an attainable target and positive prices. If the Hicksian minimiser is unique, expenditure is differentiable in prices and

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

Equivalently, wherever e is differentiable, every minimiser equals $\nabla_p e$.

Interpretation: a small price increase costs the compensated consumer approximately $h_i \, dp_i$.

Recover quantities from cost

Cobb–Douglas: $e = 2\bar{u}\sqrt{p_1 p_2}$.

$$e_{p_1} = \bar{u}\sqrt{p_2/p_1} = h_1, \qquad e_{p_2} = \bar{u}\sqrt{p_1/p_2} = h_2.$$

Complements: $e = \bar{u}(p_1 + p_2)$, so $h_1 = h_2 = \bar{u}$.

Substitutes: $e = \bar{u}\min\{p_1, p_2\}$; differentiate only within a strict price regime.

The compensated law of demand

Fix $\bar{u}$. Compare two price vectors $p^1, p^2 \gg 0$ and choose

$$h^1 \in H(p^1, \bar{u}), \qquad h^2 \in H(p^2, \bar{u}).$$

$$(p^2 - p^1) \cdot (h^2 - h^1) \leq 0.$$

If only the price of good i rises, then $h_i^2 \leq h_i^1$.

The utility target is the same in both choices; no differentiability is needed.

Two cost comparisons prove the law

h^1 is cheapest at p^1: $p^1 \cdot h^1 \leq p^1 \cdot h^2$.

h^2 is cheapest at p^2: $p^2 \cdot h^2 \leq p^2 \cdot h^1$.

Add and rearrange:

$$p^1 \cdot h^1 + p^2 \cdot h^2 \leq p^1 \cdot h^2 + p^2 \cdot h^1 \quad \Rightarrow \quad (p^2 - p^1) \cdot (h^2 - h^1) \leq 0.$$

For differentiable unique demand, $\partial h_i / \partial p_i \leq 0$.

Symmetry of compensated price effects

Suppose $e(\cdot, \bar{u})$ is C^2 locally. Then

$$\frac{\partial h_i}{\partial p_j}(p, \bar{u}) = \frac{\partial h_j}{\partial p_i}(p, \bar{u}).$$

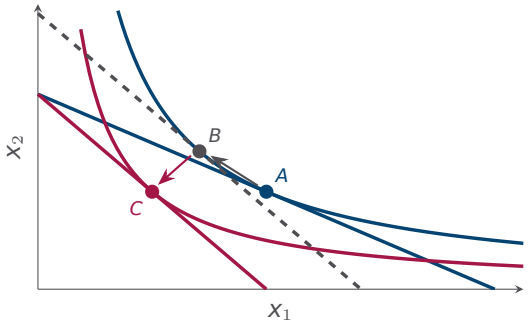
By Shephard's lemma, $h_i = e_{p_i}$ and $h_j = e_{p_j}$.

Mixed partials commute:

$$\frac{\partial h_i}{\partial p_j} = e_{p_i p_j} = e_{p_j p_i} = \frac{\partial h_j}{\partial p_i}.$$

Equivalently, $S_{ij} := \partial h_i / \partial p_j$ defines a symmetric matrix.

Substitution and income effects



A: original choice at prices p^0 and income m.
Raise p_1, keeping income fixed.
C: new choice, lower utility.
B: new prices, old utility.
$A \to B$: substitution effect.
$B \to C$: income effect.

The choices match. Their price derivatives do not.

$$h(p, \bar{u}) = x(p, e(p, \bar{u})).$$

On the left, change prices while holding $\bar{u}$ fixed.

On the right, change prices *and adjust income* to $e(p, \bar{u})$.

Which two terms appear when we differentiate the right-hand side?

Slutsky: the chain rule does the work

Assume unique demands with the derivatives below. Start from

$$h_i(p, \bar{u}) = x_i(p, e(p, \bar{u})).$$

Differentiate in p_j, holding $\bar{u}$ fixed:

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial m}\frac{\partial e}{\partial p_j}.$$

Shephard gives $e_{p_j} = h_j$. At $\bar{u} = v(p, m)$, $h_j = x_j$ and $e = m$.

The Slutsky equation

At $\bar{u} = v(p, m)$:

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total price effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{purchasing-power term}} .$$

A small increase dp_j makes the original bundle cost $x_j\, dp_j$ more.

Compensation removes that loss. Without compensation, the income response remains.

The quantity multiplying the income derivative is x_j, not x_i.

When must ordinary demand slope down?

For the own price:

$$\frac{\partial x_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\leq 0} - x_i \frac{\partial x_i}{\partial m}.$$

Normal good: $\partial x_i / \partial m \geq 0$; both effects are weakly nonpositive.

Inferior good: $\partial x_i / \partial m < 0$; the income effect is positive.

Giffen good: $\partial x_i / \partial p_i > 0$; the positive income effect dominates.

Cobb–Douglas: split an own-price effect

NES

$$x_1 = \frac{m}{2p_1}, \qquad h_1 = \bar{u}\sqrt{p_2/p_1}.$$

At the matching target,

$$\frac{\partial h_1}{\partial p_1} = -\frac{m}{4p_1^2}, \qquad -x_1\frac{\partial x_1}{\partial m} = -\frac{m}{4p_1^2}.$$

$$\frac{\partial x_1}{\partial p_1} = -\frac{m}{2p_1^2}.$$

At $p = (1, 1)$, $m = 8$: total -4 = substitution -2 + income -2.

No total effect does not mean no substitution

Cobb–Douglas gives $x_1 = m/(2p_1)$, so $\partial x_1/\partial p_2 = 0$.

Does good 2 have no effect on the demand for good 1?

$$\frac{\partial h_1}{\partial p_2} = \frac{m}{4p_1p_2}, \qquad -x_2\frac{\partial x_1}{\partial m} = -\frac{m}{4p_1p_2}.$$

The positive compensated substitution effect and negative income effect cancel exactly.

At $p = (1, 1)$, $m = 8$: $0 = 2 - 2$.

Perfect complements: all income, no substitution

For $u = \min\{x_1, x_2\}$:

$$x_1 = x_2 = \frac{m}{p_1 + p_2}, \qquad h_1 = h_2 = \bar{u}.$$

All compensated price derivatives are zero.

For either i, j,

$$\frac{\partial x_i}{\partial p_j} = -\frac{m}{(p_1 + p_2)^2} = 0 - \frac{m}{p_1 + p_2}\frac{1}{p_1 + p_2}.$$

Ordinary demands fall together even though compensated quantities stay fixed.

Quasilinearity: income effects depend on the regime

$u(a, q) = a + 4\log(1 + q)$, $a + rq \leq m$, $a, q \geq 0$.

Both goods: if $r < 4$ and $m > 4 - r$,

$$q^* = 4/r - 1, \qquad q_m = 0, \qquad q_r = h_{q,r} = -4/r^2.$$

All income on q: if $r < 4$ and $m < 4 - r$,

$$q^* = m/r, \qquad q_m = 1/r, \qquad q_r = -m/r^2, \qquad h_{q,r} = 0.$$

“No income effect” applies within the interior numeraire regime.

Can a higher price make you buy more?

NES

At fixed tastes and income, a Giffen response means $\partial x_i / \partial p_i > 0$.

It requires an inferior good and a large enough income effect.

Prestige, changed quality beliefs and changing tastes are different mechanisms.

Can we construct a Giffen good with continuous, monotone, convex preferences?

Yes. A smooth utility function is not needed for a smooth local demand.

The classic potato story

- A poor household lives mostly on potatoes, with a little meat.
- Potatoes become more expensive. The same food budget now buys less.
- The household cuts meat and replaces those calories with potatoes, still the cheapest source of calories.
- Potato purchases can rise: the loss of purchasing power outweighs substitution away from potatoes.

A subsistence kink generates Giffen demand

Let

$$u(x) = \min\{x_1 + x_2,\ 3x_1 + 2x_2 - 10\}.$$

At $1 < p_1/p_2 < 3/2$ and $5p_1 < m < 10p_2$, the unique optimum satisfies

$$2x_1 + x_2 = 10, \qquad p_1x_1 + p_2x_2 = m.$$

Hence

$$x_1 = \frac{10p_2 - m}{2p_2 - p_1}, \qquad x_2 = \frac{2m - 10p_1}{2p_2 - p_1}.$$

The utility is continuous, strongly increasing and concave.

The income effect wins

Fix $p_2 = 1$, $m = 8$. When $p_1 = 1.2$, $x = (2.5, 5)$.

Raise p_1 to 1.3: $x = (20/7, 30/7)$. Good 1 consumption rises.

Locally,

$$\frac{\partial x_1}{\partial m} = -\frac{1}{2p_2 - p_1} < 0, \qquad \frac{\partial x_1}{\partial p_1} = \frac{10p_2 - m}{(2p_2 - p_1)^2} > 0.$$

At the original point, $h_{1,p_1} = 0$ and Slutsky gives $3.125 = 0 - (2.5)(-1.25)$.

Checkpoint

- At one point, $x_1 = 2$, $x_{1,m} = -1$ and $h_{1,p_1} = -3$. Is good 1 Giffen?
- If $x_{1,p_2} = 0$, must $h_{1,p_2} = 0$?
- Can raising only p_1 increase h_1 at a fixed utility target?
- With quasilinear utility, can a low-income consumer's demand for the nonlinear good respond to income?

Substitutes and complements: specify the experiment

For $i \neq j$, **net** substitution uses $\partial h_i / \partial p_j$: positive means net substitutes; negative means net complements.

Gross substitution uses $\partial x_i / \partial p_j$ at fixed income.

Net effects are symmetric under the smooth assumptions. Gross effects need not be.

Leontief goods have zero net cross-effects and negative gross cross-effects.

A surprising restriction with only two goods

NES

For two goods, the first row of $Sp = 0$ says

$$S_{11}p_1 + S_{12}p_2 = 0.$$

Since $S_{11} \leq 0$ and prices are positive,

$$S_{12} = -S_{11}\frac{p_1}{p_2} \geq 0.$$

Two goods cannot be strict net complements in this setting. Zero is possible.

With three or more goods, negative net cross-effects are possible.

Construct compensated effects from ordinary demand

Slutsky can be rearranged:

$$S_{ij} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial m} x_j.$$

In matrix form,

$$S = D_p x + (D_m x) x^\top.$$

The left side is compensated. Every object on the right concerns ordinary choices.

Test: $S = S^\top$, $z^\top S z \leq 0$, $Sp = 0$, plus homogeneity and Walras.

Cobb–Douglas passes the matrix test

NES

For $x_i = m/(2p_i)$,

$$S = \frac{m}{4}\begin{pmatrix} -1/p_1^2 & 1/(p_1p_2) \\ 1/(p_1p_2) & -1/p_2^2 \end{pmatrix}.$$

It is symmetric and $Sp = 0$.

For every z,

$$z^\top Sz = -\frac{m}{4}\left(\frac{z_1}{p_1} - \frac{z_2}{p_2}\right)^2 \le 0.$$

At $p = (1,1)$, $m = 8$: $S = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$.

Could these choices come from our consumer model?

NES

Suppose estimated demand is

$$x_1(p, m) = \frac{m p_1}{p_1^2 + p_2^2}, \qquad x_2(p, m) = \frac{m p_2}{p_1^2 + p_2^2}.$$

It is positive, smooth, homogeneous of degree 0, and $p \cdot x = m$.

At $p = (1, 1)$ and $m = 2$, calculate the Slutsky matrix. Does the model survive?

Start with $x = (1, 1)$ and $D_m x = (1/2, 1/2)^\top$.

A symmetric matrix can still fail

NES

At $p = (1, 1)$, $m = 2$,

$$D_p x = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

S is symmetric and $Sp = 0$. But $S_{11} = 1/2 > 0$.

With $z = (1, -1)^\top$, $z^\top S z = 2 > 0$.

The demand violates compensated demand restrictions. It cannot be generated by the maintained consumer model.

Checkpoint

- Strictly convex preferences imply strictly concave expenditure in prices.
- A compensated own-price effect must be strictly negative.
- Gross cross-price effects must be symmetric.
- A Giffen good with positive consumption must be inferior at that point.

One structure, several ways to use it

NES

$$u \longrightarrow v,\ x \qquad \longleftrightarrow \qquad e,\ h.$$

Roy: differentiate indirect utility to recover ordinary demand.

Shephard: differentiate expenditure to recover compensated demand.

Slutsky: connect the two responses and obtain restrictions we can test.

Read next

Varian, Intermediate Microeconomics, 9th edition:

Chapter 8, *Slutsky Equation*.

Optional: Mas-Colell, Whinston and Green (1995), Chapter 3, *Classical Demand Theory*: expenditure minimisation, duality and integrability.

For monetary welfare measures: Varian, Chapter 14, *Consumer's Surplus*.

Homework 1 covers Lectures 1–2 and is due **14 November 2026**.

What can only increase expenditure?

If $q \geq p$, then $e(q, \bar{u}) \geq e(p, \bar{u})$.

Proof. Every target-feasible bundle costs weakly more at q.

If $u(0) \leq \bar{u} < \bar{u}'$ are attainable, then $e(p, \bar{u}) < e(p, \bar{u}')$.

Otherwise a minimiser for $\bar{u}'$ would also minimise cost for $\bar{u}$, contradicting no excess utility.

Expenditure is concave in prices

Proposition. For a fixed attainable target and $0 \leq t \leq 1$,

$$e(tp + (1-t)q, \bar{u}) \geq t\,e(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

No convexity of preferences is needed.

Why does minimising linear functions produce this direction of the inequality?

The same bundle must be purchased at the mixed prices. Reoptimising separately can only be cheaper.

Concavity: one bundle, two price vectors

Pick $z \in H(tp + (1-t)q, \bar{u})$.

Since z attains the target,

$$p \cdot z \geq e(p, \bar{u}), \qquad q \cdot z \geq e(q, \bar{u}).$$

Therefore

$$\begin{aligned} e(tp + (1-t)q, \bar{u}) &= (tp + (1-t)q) \cdot z \\ &\geq t\, e(p, \bar{u}) + (1-t) e(q, \bar{u}). \end{aligned}$$

Utility labels can change; the cost cannot

Let $\widetilde{u} = F \circ u$ for a strictly increasing F.

$$\{x : \widetilde{u}(x) \geq F(\bar{u})\} = \{x : u(x) \geq \bar{u}\}.$$

So $\widetilde{e}(p, F(\bar{u})) = e(p, \bar{u})$ and $\widetilde{H}(p, F(\bar{u})) = H(p, \bar{u})$.

With continuous LNS utility, e is continuous in positive prices and interior attainable targets.

Why the quantity appears in the derivative

Let $h = h(p, \bar{u})$ and choose $h_t \in H(p + te_i, \bar{u})$, where e_i is the ith unit vector.

For $t > 0$, using each bundle in the other problem gives

$$t(h_t)_i \;\leq\; e(p + te_i, \bar{u}) - e(p, \bar{u}) \;\leq\; th_i.$$

Uniqueness and local compactness give $h_t \to h$ as $t \to 0$.

Divide by t and take limits; negative t gives the same derivative h_i.

What if demand is not unique?

NES

For any $h \in H(p, \bar{u})$ and any $q \gg 0$,

$$e(q, \bar{u}) \leq q \cdot h = e(p, \bar{u}) + (q - p) \cdot h.$$

If e is differentiable, this supporting-plane inequality forces $h = \nabla_p e(p, \bar{u})$.

For $u = x_1 + x_2$, equal prices give many minimisers and a kink in e.

Use the demand correspondence at a kink; a single derivative cannot select the whole set.

The matrix behind compensated substitution

Assume $e(\cdot, \bar{u})$ is C^2 locally. Define

$$S_{ij} = \frac{\partial h_i}{\partial p_j}, \qquad S = D_p h = D_{pp}^2 e.$$

Symmetry: $S_{ij} = S_{ji}$.

Negative semidefiniteness: $z^\top S z \leq 0$ for every $z \in \mathbb{R}^n$.

In particular, $S_{ii} \leq 0$. Zero effects are allowed.

Why symmetry and the quadratic inequality?

By Shephard, $h_i = e_{p_i}$. Mixed partials of a C^2 function commute:

$$\frac{\partial h_i}{\partial p_j} = e_{p_i p_j} = e_{p_j p_i} = \frac{\partial h_j}{\partial p_i}.$$

For any direction z, define $g(t) = e(p + tz, \bar{u})$ near $t = 0$.

Concavity of e implies concavity of g, hence

$$g''(0) = z^\top D^2_{pp} e(p, \bar{u}) z \leq 0.$$

A common price increase changes no compensated quantities

NES

Recall $h(tp, \bar{u}) = h(p, \bar{u})$. Differentiate in t at $t = 1$:

$$\sum_j \frac{\partial h_i}{\partial p_j} p_j = 0 \quad \text{for every } i, \qquad Sp = 0.$$

Since S is symmetric, also $p^\top S = 0$.

Relative prices drive compensated substitution; a change of currency does not.

Reserve: compensation and equivalent variation

For a price increase $p^1 \geq p^0$ at fixed income m, set $u^0 = v(p^0, m)$ and $u^1 = v(p^1, m)$.

Compensating variation, as a positive loss:

$$CV = e(p^1, u^0) - m.$$

Equivalent variation, as a positive loss:

$$EV = m - e(p^0, u^1).$$

CV restores old utility at new prices; EV removes the equivalent income at old prices.

Reserve: how much compensation?

For $u = \sqrt{x_1 x_2}$, $p^0 = (1, 1)$, $p^1 = (2, 1)$ and $m = 8$:

$$u^0 = 4, \qquad u^1 = 2\sqrt{2}.$$

$$CV = 8\sqrt{2} - 8 \approx 3.314.$$

$$EV = 8 - 4\sqrt{2} \approx 2.343.$$

These are exact finite-change measures; a derivative is only a local approximation.

Reserve: two meanings of compensation

Hicks compensation: hold old utility u^0 fixed; new income is $e(p^1, u^0)$.

Slutsky compensation: keep the old bundle x^0 affordable; new income is $p^1 \cdot x^0$.

For finite price changes these generally differ. In our example: $8\sqrt{2}$ versus 12.

To first order they agree: $de = h(p^0, u^0) \cdot dp = x^0 \cdot dp$.

Reserve: can we work backwards from demand?

NES

A rationalisable smooth demand must satisfy homogeneity, Walras and a symmetric negative semidefinite Slutsky matrix.

Normalize utility at reference prices p^0 so that $e(p^0, \bar{u}) = \bar{u}$.

Then expenditure must solve

$$\nabla_p e(p, \bar{u}) = x(p, e(p, \bar{u})).$$

This is the integrability problem. Converse theorems require additional domain and regularity conditions.

Reserve: genuine net complements need another good

Let $u(x) = \log(\min\{x_1, x_2\}) + x_3$, with $x_1, x_2 > 0$, $x_3 \geq 0$.

Write $c = p_1 + p_2$. In the regime $\bar{u} > \log(p_3/c)$,

$$h_1 = h_2 = \frac{p_3}{c}, \qquad h_3 = \bar{u} - \log(p_3/c) > 0.$$

$e = p_3\big[1 + \bar{u} - \log p_3 + \log(p_1 + p_2)\big]$.

Goods 1 and 2 are net complements:

$$\frac{\partial h_1}{\partial p_2} = -\frac{p_3}{(p_1 + p_2)^2} < 0.$$

Reserve: continuity of expenditure

Take $(p^k, \bar{u}^k) \to (p, \bar{u})$, with $p \gg 0$ and an interior attainable target.

LNS gives a bundle z arbitrarily close to an optimum h, with $u(z) > \bar{u}$. It is feasible for all large k.

Thus $\limsup e(p^k, \bar{u}^k) \leq p \cdot z$; let $z \to h$ to get the upper bound $e(p, \bar{u})$.

Those cost bounds make minimisers locally bounded. A convergent subsequence is target-feasible by continuity, giving $\liminf e(p^k, \bar{u}^k) \geq e(p, \bar{u})$.